

The optimality conditions for quadratic functions are related to the positive definiteness of the (constant) second derivative matrix. The derivatives of the objective function are

$$\nabla f(x) = Qx + g, \quad \nabla^2 f(x) = Q.$$

The positive definiteness of Q can be identified from its eigenvalues, that are the roots of the characteristic polynomial. Therefore, we solve the equation

$$\begin{aligned} \det(\lambda I - Q) &= \det \begin{pmatrix} 1 - \lambda & \alpha \\ \alpha & 1 - \lambda \end{pmatrix} = (1 - \lambda)^2 - \alpha^2 \\ &= \lambda^2 - 2\lambda + (1 - \alpha^2) \\ &= (1 - \alpha - \lambda)(1 + \alpha - \lambda). \end{aligned}$$

We obtain the eigenvalues:

$$\begin{aligned} \lambda_1 &= 1 - \alpha \\ \lambda_2 &= 1 + \alpha. \end{aligned}$$

The corresponding (normalized) eigenvectors are

$$\begin{aligned} u_1 &= \begin{pmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{pmatrix} & \text{as } Qu_1 &= \lambda_1 u_1, \\ u_2 &= \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} & \text{as } Qu_2 &= \lambda_2 u_2. \end{aligned}$$

We first investigate the optimality conditions.

Necessary conditions When Q is not positive semidefinite, that is, when at least one eigenvalue is non positive, there is no optimal solution. This is the case if $\alpha > 1$ (so that $\lambda_1 < 0$), or $\alpha < -1$ (so that $\lambda_2 < 0$).

Sufficient conditions When Q is positive definite, the problem has a unique global minimum. This is the case if $-1 < \alpha < 1$, so that $\lambda_1 > 0$ and $\lambda_2 > 0$.

Now, what happens when Q is positive semidefinite (so that the necessary conditions are verified) but not positive definite (so that the sufficient conditions are not verified)?

This is the case if $\alpha = 1$, so that $\lambda_1 = 0$ and $\lambda_2 = 2$, or if $\alpha = -1$, so that $\lambda_1 = 2$ and $\lambda_2 = 0$.

Geometrically, we can decompose the space into two subspaces. In the subspace spanned by the eigenvectors of the positive eigenvalues, the function is strictly convex. In this subspace, there is a unique minimum. In the subspace spanned by the eigenvectors of the zero eigenvalues, the function is linear, as there is no curvature. Therefore, it is bounded only if it is constant. And, in that case, there is an infinite number of minima.

Algebraically, we use the Schur decomposition of Q :

$$\begin{aligned} Q &= U\Lambda U^T \\ &= \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 1 - \alpha & 0 \\ 0 & 1 + \alpha \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}, \end{aligned}$$

where $\mathbf{U}$ is an orthogonal matrix composed of the eigenvectors of $\mathbf{Q}$ organized in columns.

Let's use the matrix $\mathbf{U}$ to decompose $\mathbf{x}$ and $\mathbf{g}$:

$$\mathbf{U}^T \mathbf{x} = \mathbf{x}' = \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} \text{ and } \mathbf{U}^T \mathbf{g} = \mathbf{g}' = \begin{pmatrix} g'_1 \\ g'_2 \end{pmatrix},$$

that is

$$\mathbf{x}' = \frac{\sqrt{2}}{2} \begin{pmatrix} x_1 - x_2 \\ x_1 + x_2 \end{pmatrix} \text{ and } \mathbf{g}' = \frac{\sqrt{2}}{2} \begin{pmatrix} g_1 - g_2 \\ g_1 + g_2 \end{pmatrix}.$$

We use the Schur decomposition, and the fact that the $\mathbf{U}$ matrix is orthogonal, so that $\mathbf{U}\mathbf{U}^T = \mathbf{I}$, to rewrite the objective function as follows:

$$\begin{aligned} f(\mathbf{x}) &= \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{g}^T \mathbf{x} + c \\ &= \frac{1}{2} \mathbf{x}^T \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T \mathbf{x} + \mathbf{g}^T \mathbf{U} \mathbf{U}^T \mathbf{x} + c. \end{aligned}$$

Using the decomposition of the vectors, we obtain the function in the new variables

$$\begin{aligned} \tilde{f}(\mathbf{x}') &= \frac{1}{2} \mathbf{x}'^T \mathbf{\Lambda} \mathbf{x}' + \mathbf{g}'^T \mathbf{x}' + c \\ &= \frac{1}{2} \lambda_1 (x'_1)^2 + \frac{1}{2} \lambda_2 (x'_2)^2 + g'_1 x'_1 + g'_2 x'_2. \end{aligned}$$

The gradient of this function is

$$\nabla \tilde{f}(\mathbf{x}') = \begin{pmatrix} \lambda_1 x'_1 + g'_1 \\ \lambda_2 x'_2 + g'_2 \end{pmatrix}.$$

We see that, for the gradient to be zero (which is a necessary condition for optimality), we need

$$\begin{aligned} x'_i &= \frac{-g'_i}{\lambda_i} & \text{if } \lambda_i \neq 0, \\ g'_i &= 0 & \text{if } \lambda_i = 0. \end{aligned}$$

Now, we consider the two values of α identified above.

- If $\alpha = 1$, then $\lambda_1 = 0$ and $\lambda_2 = 2$, the gradient is zero when

$$g'_1 = \frac{\sqrt{2}}{2} (g_1 - g_2) = 0 \text{ and } x'_2 = \frac{\sqrt{2}}{2} (x_1 + x_2) = \frac{-g'_2}{2}.$$

It means that the function must be such that $g_1 = g_2$. If so, any solution such that

$$x_1 = -g_1 - x_2$$

is a global minimum. It can easily be verified that the gradient of the function

$$f(\mathbf{x}) = \mathbf{Q} \mathbf{x} + \mathbf{g} = \begin{pmatrix} x_1 + \alpha x_2 + g_1 \\ \alpha x_1 + x_2 + g_2 \end{pmatrix}$$

is indeed zero when $\alpha = 1$, $x_1 = -g_1 - x_2$ and $g_1 = g_2$, for any value of x_2 .

If $g_1 \neq g_2$, the problem is unbounded and has no optimal solution.

- If $\alpha = -1$, then $\lambda_1 = 2$ and $\lambda_2 = 0$, and we need

$$x_1' = \frac{\sqrt{2}}{2}(x_1 - x_2) = \frac{-g_1'}{2} \text{ and } g_2' = \frac{\sqrt{2}}{2}(g_1 + g_2) = 0.$$

It means that the function must be such that $g_1 = -g_2$. If so, any solution such that

$$x_1 = -g_1 + x_2$$

is a global minimum. It can easily be verified that the gradient of the function

$$f(x) = Qx + g = \begin{pmatrix} x_1 + \alpha x_2 + g_1 \\ \alpha x_1 + x_2 + g_2 \end{pmatrix}$$

is indeed zero when $\alpha = -1$, $x_1 = -g_1 + x_2$ and $g_1 = -g_2$, for any value of x_2 .

If $g_1 \neq -g_2$, the problem is unbounded and has no optimal solution.

We can now answer the questions.

1. For which values of α f does not have a local minimum?
 - If $\alpha > 1$, so that $\lambda_1 < 0$ (from the necessary conditions),
 - if $\alpha < -1$, so that $\lambda_2 < 0$ (from the necessary conditions),
 - if $\alpha = 1$ when $g_1 \neq g_2$ (from the discussion above),
 - if $\alpha = -1$ when $g_1 \neq -g_2$ (from the discussion above).
2. For which values of α f has a unique global minimum? If $-1 < \alpha < 1$, so that $\lambda_1 > 0$ and $\lambda_2 > 0$ (from the sufficient conditions).
3. What are the conditions on g and α so that the problem has an infinite number of global minima? From the discussion above,
 - if $\alpha = 1$ when $g_1 = g_2$,
 - if $\alpha = -1$ when $g_1 = -g_2$.